Unlocking Automation: Core Concepts in n8n

This presentation simplifies the fundamental building blocks of n8n, using relatable analogies and practical examples. We'll demystify workflows, nodes, triggers, and more, culminating in a comprehensive summary to solidify your understanding.

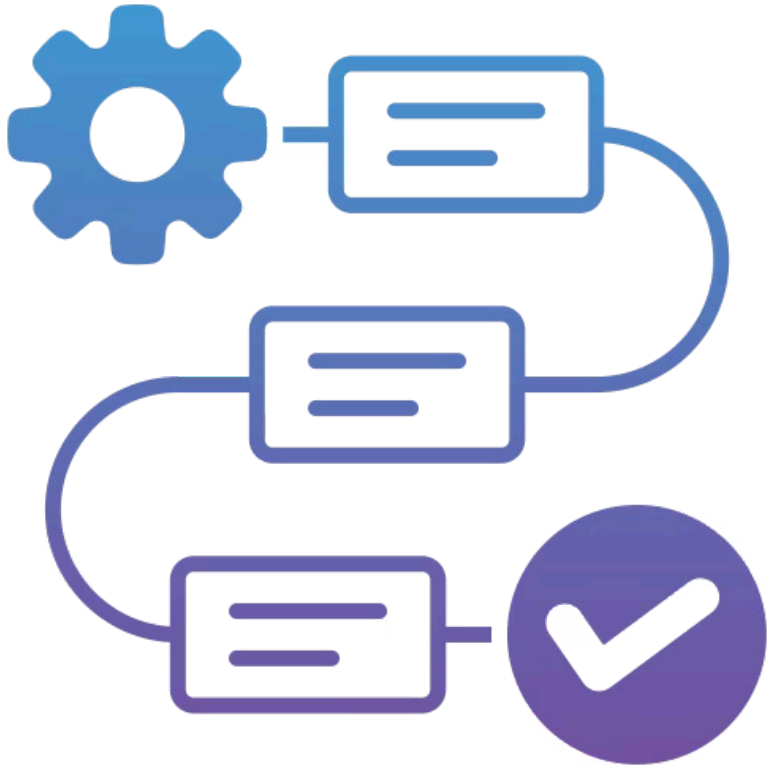
Made with GAMMA

Workflows: Your Automation Blueprints

Think of a **workflow** as a *recipe* or a *project plan*. Just like a recipe has ingredients and steps, a workflow has **nodes** connected step-by-step.

- **Analogy:** A company's "Lead Management" workflow might trigger on a new form submission, enrich data, send a welcome email, and add the lead to CRM. It's one complete, automated process.
- Your "morning routine" is a workflow: trigger "wake up" leads to steps like brush teeth, make coffee, and check calendar.

> **Key Idea:** A workflow equals one complete automation journey.



Made with GAMMA

Nodes: The Individual Steps

A **node** is a single, atomic step within your workflow. Each node performs a specific job, like *sending an email*, *fetching data*, or *checking a condition*.

- **Analogy:** Imagine a factory assembly line. Each worker (node) specialises in one task: painting, quality control, or packaging.
- **Professional Example:** A "Gmail Send" node's job is to compose and send a thank-you email automatically.
- **Personal Example:** A "Calendar Create" node's job is to add "Dinner with friend" to your schedule.

Made with GAMMA

Triggers: Initiating the Process

Triggers are the events that kick off your workflows. They are the initial spark that sets the automation in motion.

- **Analogy:** Think of a **doorbell** – it rings, and that sound immediately initiates a series of actions, like opening the door or checking who's there.
- **Examples:**
  - **Manual Trigger:** You explicitly press a button to start (like testing a new machine).
  - **Schedule Trigger:** An alarm clock, reliably going off at a fixed time every day.
  - **Webhook Trigger:** A delivery app pinging your phone the moment your food is ready.
  - **Telegram/Gmail Trigger:** The workflow starts automatically when you receive a specific message or email.

Made with GAMMA

Actions: The Workflow's Purpose

Actions are the subsequent tasks your workflow performs once a trigger has been activated. They are the actual "doing" part of your automation.

- **Analogy:** After your alarm rings (the trigger), the actions are what you actually *do* next: getting out of bed, making coffee, getting dressed.
- **Example:** Upon a new form submission (trigger), an action could be to send a detailed Slack notification to your team, ensuring immediate awareness.
- **Example :** If you receive a specific Telegram message (trigger), an action could be to fetch your Google Calendar and reply automatically with your current availability.

Made with GAMMA

Data Flow: The Information Highway

Data flow describes how information, the **fuel** of your automation, seamlessly moves and transforms between nodes. Each node processes and then passes a JSON object (like a parcel) to the next node in the sequence.

- **Analogy:** Envision a relay race where each runner (node) precisely passes a baton (data) to the next runner. The baton carries crucial information forward.
- **Example JSON Item:**

```
{ "name": "Priya", "email": "priya@acme.com" }
```

- **Expression Examples (to access data):**
  - `{{$json["email"]}}` → extracts the email from the current item.
  - `{{$node["Lookup"].json["id"]}}` → retrieves an 'id' from a previous "Lookup" node's output.
- **Professional Scenario:** An API sends a customer ID; this ID is then passed to a subsequent node to fetch the customer's full details from a database.
- **Personal Scenario:** A Telegram message contains "movie time 7pm"; a later node uses this specific time to add a calendar event.

Made with GAMMA

JSON Basics: The Workflow's Blueprint

Your n8n workflows are fundamentally saved and shared as **JSON files**. JSON (JavaScript Object Notation) is a lightweight data-interchange format, structured yet human-readable.

- **Analogy:** Consider JSON as the *blueprint* of your house. Every detail – walls, windows, doors, their precise locations and specifications – is meticulously documented within this blueprint.
- You don't need to "speak JSON" fluently for daily n8n use, but understanding its structure is incredibly helpful when exporting, importing, or debugging complex workflows.
- **Example Workflow JSON (simplified):**

```
{
 "name": "My Workflow",
 "nodes": [
 {
 "name": "Webhook Trigger",
 "type": "webhook"
 },
 {
 "name": "Gmail Send",
 "type": "gmail"
 }
 ],
 "connections": {
 "Webhook Trigger": ["Gmail Send"]
 }
}
```

Made with GAMMA

Recap

1

Workflow

Recipe / Project Plan

2

Node

Worker in a factory line / Step in a recipe

3

Trigger

Doorbell or Alarm clock

4

Action

Cooking step or Sending an email

5

Data Flow

Relay race baton / Parcel on conveyor belt

6

JSON

Blueprint or Recipe card written down

Made with GAMMA

Summary Table: n8n Core Concepts

Topic	Explanation	Analogy	Professional Example	Personal Analogy
Workflow	Complete automation	Recipe / Project plan	Lead workflow: form → enrich → email	Morning routine: wake up → coffee
Node	Single step	Worker in factory / Recipe step	Gmail node → sends thank-you email	Calendar node → create dinner event
Trigger	Workflow starter	Doorbell / Alarm clock	Webhook: new form submission	Telegram Trigger: friend sends msg
Action	Work performed after trigger	Cooking step	Send Slack alert after form fill	Reply to Telegram with your schedule
Data Flow	Info moving between nodes	Baton in relay / Parcel flow	API ID → fetch customer details	“7pm movie” → create calendar event
JSON	Workflow blueprint format	House blueprint / Recipe card	Export workflow as JSON	Import a ready-made workflow

Made with GAMMA